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Presurgical seizure frequency and tumoral etiology predict the outcome after extratemporal epilepsy surgery

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Abstract *Objective* To examine the predictive value of demographic data for the seizure outcome after extratemporal epilepsy surgery. *Methods* Eightyone patients who underwent resective extratemporal epilepsy surgery were retrospectively studied concerning (a) age at surgery, (b) onset of epilepsy, (c) duration of epilepsy, (d) number of seizures at the time of presurgical evaluation, (d) number of presurgically tested antiepileptic substances and (f) number of seizure types. The data were correlated to the postoperative seizure outcome after two years. *Results* 33 patients (40.7%) were seizure free two years after surgery. Univariate and multivariate analysis revealed that both tumor etiology and low presurgical seizure frequency were independently associated with seizure freedom after epilepsy surgery. The recurrence rate in patients with one or more seizures per day was more than two-fold if compared with patients with fewer seizures. The remaining demographic factors did not show a significant association with seizure outcome in our 81 patients. *Conclusions* Fewer than daily seizures prior to surgery and a tumoral etiology independently increase the likelihood of remaining seizure free two years after extratemporal epilepsy surgery.

Key words epilepsy surgery · tumors · seizure frequency · demographic data · predictors for prognosis

Introduction

Resective epilepsy surgery is an elective procedure to remove or functionally disconnect the epileptogenic zone in patients with medically refractory partial epilepsies in order to prevent the onset or propagation of epileptic activity. Resections in the extratemporal cortex are performed relatively seldom and the majority of studies concerning predictors for the postoperative seizure outcome focus on temporal lobe procedures [3, 10].

Several clinical and diagnostic variables such as etiology [10], EEG data [10, 12], MRI data [13], EEG/MRI concordance [13] or an extended surgical resection [13] have a significant influence on the surgical outcome. In extratemporal epilepsy surgery, the majority of data were obtained from mixed series containing temporal and extratemporal lobe resections [3, 11] and mostly include small numbers of treated patients [9, 11]. Our group recently published the prognostic value of electrophysiological, histological and semiological data in frontal lobe and posterior cortical epilepsies respectively [2, 8].

A few studies have additionally investigated the predictive value of demographic data from the patients' history that may indicate the severity of the patients' epilepsy. However, the impact of those presurgical variables on the surgical outcome is not unequivocally clear, as data from the literature are often conflicting [10]. One reason for this may be different lesion sites and etiologies of the epilepsy, often determining the course and severity of the disease as well as the surgical outcome after corticectomies. Therefore, investigations on the heterogeneous group of neocortical epilepsy surgery should principally be separated from temporal resections as the latter are regularly characterized by more standardized surgical procedures, similar etiologies (mostly mesial temporal sclerosis) and finally a more favorable outcome.

In this study we investigated the impact of demographic data on the seizure outcome after extratemporal epilepsy surgery and correlated the data to the different histologic etiologies.

Patients and methods

We retrospectively studied 81 patients (51 male) with lesional focal epilepsies of the extratemporal cortex. All patients were evaluated for epilepsy surgery between 1991 and 2001. Resistance against at least two antiepileptic first line drugs had been established in all patients. All patients underwent video and EEG monitoring with scalp EEG. Additional invasive EEG using subdural grids, epidural electrodes or depth electrodes was applied in 66 cases. In patients with left hemispheric epilepsy (n = 36), language lateralization was performed by Wada test or electrical cortical stimulation. All patients had defined lesions in presurgical MRI that could histologically be identified as gangliogliomas (n = 6), oligodendrogliomas (n = 2), dysembryoblastic neuroepithelial tumors (DNET; n = 7), low-grade astrocytomas (n = 3), malformations of cortical development (MCD; n = 26), vascular lesions (arteriovenous malformations, cavernoma etc.; n = 8) and cortical gliosis (n = 29).

Corticectomy was performed in 80 patients whereas one patient received multiple subpial transsections (MST). In 39 patients the epileptigenic lesion was located posterior (occipital, parietal or in the temporo-parieto-occipital border) and in 42 patients in the frontal cortex respectively. Detailed information about the patients' diagnostic results can be obtained elsewhere [2, 8].

We divided the patients into two groups according to their surgical outcome — those who had no seizure at all (except auras) following surgery (Engel Ia and Ib), and those with persisting seizures (Engel Ic to IV). The surgical outcome of all patients was assessed after 2 years using Engel's classification [5].

The following demographic data were correlated with the surgical outcome: Age at surgery, onset and duration of epilepsy, the number of seizures at the time of presurgical evaluation, the number of antiepileptic substances tested presurgically, and the number of seizure types.

In this context, the term *seizure frequency* reflects the seizure load per year, calculated by multiplying the daily frequency by 365, the weekly frequency by 52 or the monthly frequency by 12. If a patient reported repetitive tonic or clonic jerks, a series was counted as one seizure whereas in a set of series with a convulsion-free interval in between, each single series was counted separately. All seizure types but auras were considered for evaluation as the auras had been documented only infrequently by the patients.

For the univariate analysis of the categorical data, chi-square tests were used. Interval- or ordinal scaled continuous data were analyzed with the Mann-Whitney U-test.

In a second step, backward stepwise multivariate logistic regression analysis was performed to determine the independent contribution of individual variables to the outcome. Seizure freedom for at least two years (Engels Class 1a/b) was coded 1 if present and 0 if absent and was considered to be the dependent variable. The other variables as mentioned above were included as independent variables: For each step, the p-value threshold for inclusion was set at 0.05, and for exclusion at 0.1. Significance level for all tests was set at $p = 0.05$ (two-tailed tests).

Results

Two years after operation, 33 patients (40.7%) were seizure-free (Engel Ia and Ib). An additional 32 patients (39.5%) benefited from the operation (Engel Ic to IIId). Sixteen patients (19.8%) had no change in seizure frequency or even worsened (Engel IV). The outcome by histologic diagnosis is given in Table 1. A tumor etiology proved to be significantly associated to a favorable outcome in both univariate and multivariate analysis (Table 2).

A summary of the demographic data and a comparison of medians between seizure free patients and those with persisting seizures are given in Table 1. The seizure-free group did not show significant differences in the variables age at surgery, epilepsy

Table 1 Pateints' demographic and histologic data with respect to the surgical outcome after two years. P valves <0.05 using Mann-Whitney-U test* and Chi-square** test

	Range	median (25/75 quartiles)			Proportion of seizure free patients	Significance
		All patients	Seizure free	Not Seizure free		
Age of surgery (years)	16–53	24.7 (19.8/33.8)	24.7	26.0		n.s. *
Epilepsy onset (years)	1–27	8.0 (3.0/12.0)	9.0	8.0		n.s. *
Duration of epilepsy (years)	2–43	19.0 (10.5/26.8)	18.3	21.0		n.s. *
Number of presurgically tested AED	2–12	5 (4.0/7.0)	5.0	5.5		n.s. *
Number of seizure types	1–3	2 (1.0/2.0)	2	1.5		n.s. *
Presurgical seizure frequency (per year)	**1–36318**	**312.0 (75.0/929.5)**	**120.0**	**548.0**		**p<0.01***
One or more seizures a. day					**11\40 (27,5%)**	**p = 0.017****
Histology						
Tumours					**11\18 (61.1%)**	**p = 0.046****
MCD					10\26 (38.5%)	n.s.**
Gliosis					10\29 (34.5%)	n.s.**
Vascular lesions					2/8 (25.0%)	n.s.**

Table 2 Multivariate analysis of the patients' demographic and histologic with respect to the surgical outcome

	Regression coefficient	Standard error	Significance
Age at surgery	0.016	0.045	n.s.
Duration of epilepsy	−0.009	0.46	n.s.
Number of presurgically tested AED	−0.004	0.101	n.s.
Number of seizure types	0.23	0.419	n.s.
More than daily seizures	**−1.205**	**0.539**	**p = 0.026**
Tumors	**1.358**	**0.673**	**p = 0.044**
MCD	0.392	0.624	n.s.
Vascular lesions	0.402	0.941	n.s.

onset, duration of epilepsy, number of anti epileptic drugs presurgically tested and number of seizure types, whereas the presurgical seizure frequency significantly differed between both groups ($p < 0.01$) with an about five-fold higher median seizure frequency in the non seizure-free group. The proportion of seizure-free patients was 53.7% (n = 22/41) if patients had less than daily seizures and was 27.5% (n = 11/40) in the group with one or more seizures per day (p = 0.017).

The median yearly seizure frequency was 456.6/year in the MCD group, 338.0/year in the vascular lesion group, 278.5/year in the tumor group and 144.0/year in the gliosis patients.

Multivariate analysis (Table 2) of the same variables revealed a low seizure frequency (p = 0.026) and tumor etiology (p = 0.044) to be correlated with a favorable outcome. Using these two variables, 45/48 patients (93.8%) with remaining post surgical seizures could be predicted correctly.

Discussion

Our retrospective study showed a non-tumoral etiology and a high presurgical seizure frequency to be independently predictive for an unfavorable outcome after resective epilepsy surgery of the extratemporal cortex. However, the presence of daily seizures does not necessarily exclude patients from epilepsy surgery as 80% of our patients suffering from daily seizures benefited from the operation (Engel Ic to IIId). The remaining demographic factors (age at surgery, onset and duration of epilepsy, number of presurgical tested antiepileptic substances and number of seizure types) showed no significant correlation to the seizure outcome 2 years after operation in our 81 patients.

Owing to the relatively low proportion of epilepsy surgery in the extratemporal cortex, reliable data about demographic predictors in this particular subgroup is scarce. In a study of 16 patients with frontal lobe epilepsies, median seizure frequency of the 10 patients with Engels Class 1 outcome after 1 year was significantly lower than of patients with a less favorable outcome [9]. In addition, the epilepsy onset tended to be later in this study. In contrast, a report of 37 tumoral frontal lobe epilepsies revealed a mild but not statistically significant lower age of epilepsy onset in patients with a Class I and II outcome (follow-up 1–15 years) [14]. However, although both studies used varying follow-up periods of at least 1 year, the definition of *favorable outcome* in terms of Engels Classification differed.

In contrast to extratemporal lobe epilepsies, the predictive value of demographic data in patients with refractory temporal lobe epilepsy (TLE) has been investigated extensively. Two systematic reviews [3, 10] summarized 33 studies focusing on presurgical demographic variables. The data showed conflicting results concerning the impact on the patients' seizure outcome in TLE and mixed series (TLE and extra-temporal lobe epilepsies) for the variables *onset of epilepsy, duration of epilepsy, age at surgery, number of seizure types and frequency of complex partial seizures*. As in our study, a significant prognostic impact of the presurgical frequency of complex partial seizures in TLE was shown in only one other study [6].

A more favorable seizure outcome after tumor resections has been shown in extra temporal epilepsies[1, 2] and temporal lobe epilepsies [4, 7]. Resections of benign cortical tumors probably have the most favorable outcome regarding seizure freedom after surgery. This may be due to the easier delineation of pathological from normal tissue, resulting in excellent surgical results after complete tumor resections compared with incomplete resections [2].

As a consequence of the limited sample size, our multivariate analysis only focused on demographic and histologic variables. Diagnostical and surgical data (e.g. semiological lateralization, EEG data or completeness of resection) which themselves showed a ignificant impact on the surgical outcome in previous studies [2, 8] had to be left out for statistical reasons. Because of this limitation future studies with larger sample sizes may be helpful.

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